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Report (Final)

**Exploring Discriminative and Efficient Data
Representation for Zero-shot Learning**

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ABSTRACT

Pre-trained vision language models such as CLIP has demonstrated remarkable performance on tasks related to analysing images, and their ability to generalize in zero-shot scenarios is particularly promising. Despite such improvements, the majority of current approaches require labelled data for all classes and this requirement may not be met in real-world applications. For instance, certain classes, like emerging concepts, might lack labelled data entirely. More notably, we can observe that even though the models normally achieve high top-5 zero-shot accuracies, they exhibit significantly lower top-1 accuracies, with a gap exceeding 25% in certain instances. One of the main reasons behind this performance disparity stem from specific cases of ambiguity in the provided text prompts. In order to tackle such problems, we are proposing an Automatic Generated Prompts system (*AGP*) that can generate features for categories lacking labelled data by leveraging generative models such as CLIP. We use a combination of prompt learning method and variational autoencoders to generate prompts for label-only classes and consequently, visual features can then be obtained by feeding these created prompts along with the class names into CLIP's language encoder. Moreover, to further improve the top-1 prediction accuracies, we leverage the WordNet hierarchy, where the original class is expanded in the form of text prompts by integrating both its children and parent using the semantic label hierarchy. Thorough experimentation on the CLIP model is conducted using distinct datasets to demonstrate the efficacy of our approach and highlight its improvement.

TABLE OF CONTENTS

1	INTRODUCTION	4
2	RELATED WORK.....	6
2.1	Generalized Zero-Shot Learning.....	6
2.2	Prompt Learning	6
2.3	Label Hierarchy	7
3	PROPOSED METHODOLOGY	8
3.1	Background.....	8
3.2	AGP: Automatic Generated Prompts for Vision-Language Models.....	8
3.3	Semantic Hierarchy Augmentation.....	11
4	EXPERIMENTS AND RESULTS.....	15
4.1	Generalized zero-shot learning	15
4.2	Enhancing ImageNet Classification with Semantic Hierarchy Augmentation	18
4.3	Ablation Study	19
5	CONCLUSION, LIMITATIONS AND FUTURE WORK.....	21
6	REFERENCES	22

1 Introduction

In the last few years, language-supervised vision pre-trained models have emerged as powerful tools, showcasing impressive zero-shot capabilities and transfer abilities [1, 2]. Among these, CLIP (Contrastive Language–Image Pre-training) [1] stands out as a large-scale vision-language model aligning textual and visual representations from a vast dataset of 400 million image-text pairs, using a language and a visual encoder.

CLIP [1] excels in zero-shot recognition, generating classification weights through prompting with class names [3]. But a major limitation of this approach is its dependence of learning from classes with available data. Real-world scenarios, following the principles of Zipf’s law, often lack data for rare categories. This limitation hinders the applicability of previous methods in cases where the number of potential classes is extensive, or where acquiring labelled data for every possible category is impractical [4, 5]. In addition to the approach stated above, another technique is also implemented in an attempt to improve the prediction accuracy of the CLIP model. An observed disparity exists in the ImageNet dataset [6], where the zero-shot top-1 accuracy stands at 63.5%, while the top-5 accuracy is notably higher at 87.1%. This discrepancy in the model's performance suggests a scope for enhancement. Upon investigation, it has been discerned that numerous top-1 errors were attributed to ambiguous text prompts, originating from ImageNet's WordNet hierarchy. For example, certain object names are overly general, leading to the model struggling to identify specific subclasses (e.g., potentially mistaking "hot-air balloons" with "airships"). Conversely, some object names are overly specific, causing the model difficulty in associating them with broader categories (e.g., confusing "tuskers" with "Asian elephants").

To address such observations, we propose to introduce a novel generative approach, Automatic Generated Prompts (AGP) system, aimed at generating features for categories without data. However, one of the significant challenges using generative models is that it typically demands a substantial amount of training data, which contradicts our aim for data efficiency. To address this, we design a Variational Autoencoder (VAE) [7] framework and train the generator to create prompts (semantic descriptions) for new, unseen classes. These prompts, along with their class names, are then used to generate some features. We also introduce a Semantic Hierarchy

Augmentation [30] approach to address issues related to ambiguous text prompts and WordNet hierarchical structures. By identifying images with potentially incorrect predictions, we estimate the prediction confidence by using different text prompts and applying various transformations to the image. For each specific change that we initialize, the prediction confidence can be calculated using common methods such as the Maximum SoftMax Probability [8]. Then, if the predictions aren't consistent, it should suggest low prediction confidence. To improve accuracy and the model robustness for these cases, we implement a labelling method which utilizes both ancestor (top-down) and children (bottom-up) labels from WordNet.

In conclusion, this paper addresses the challenge in cases where some classes lack relevant data, all while maintaining a high level of top-1 and top-5 prediction accuracies. Through extensive experiments encompassing generalized zero-shot learning, identifying failure modes in zero-shot ImageNet classification, emphasizing the sensitivity of the text encoder to prompts, and proposing a refined zero-shot confidence score for multi-modal models, our approach aims to significantly improve prediction accuracies for new classes and the low confidence subset of images.

2 Related Work

2.1 Generalized Zero-Shot Learning

Our project's objectives align with the field of Generalized Zero-Shot Learning (GZSL), which focuses on enabling recognition of objects from both known (seen) and unknown (unseen) classes. This involves training models capable of identifying objects, even objects which were not part of the data that was used to initially train the model, and it also typically employs extra semantic information like attributes or text descriptions [9] for both known and unknown classes. In GZSL, there are two main approaches: embedding-based and generative-based methods. Embedding-based methods aim to connect visual features to their semantic counterparts [10, 11], but they often show bias towards known classes. To address this issue, GZSL methods have been developed with generative abilities. These methods work on creating visual features for unseen classes through techniques like Variational Autoencoders (VAEs) [7, 12] or Generative Adversarial Networks (GANs) [12, 13, 14]. However, one of the major challenges with these generative methods is that they typically need a lot of labelled data for known classes in order to learn how to create visual features for unseen classes. In our project, we seek to replicate the idea of GZSL by figuring out how to create new samples for such classes. The challenge that we face is that we don't have much labelled data in our training set, and the descriptions we have for classes are quite basic, such as "a photo of a {class}.", and because of these limitations, the existing GZSL generative methods struggle to generate accurate new samples for these unseen classes.

2.2 Prompt Learning

Prompt learning tools are now very significant, especially with the development of image-text models [15, 16]. These models align images with their textual descriptions, making it easier to generalize to unseen categories during testing. However, it's been noticed that the accuracy of image classification in downstream tasks depends heavily on the specific input prompts given to the model. This has led researchers to explore ways to improve the performance of cross-modal models in various image classification tasks by adjusting the prompts provided to such models. Recent methods like Context Optimization (CoOp) [6, 17] enhance the performance of recognizing new classes by learning prompts that are customized for each specific image. These prompts are generated based on the image itself, and this approach allows for the easy transfer of

prompts to identify new class samples. Unlike [18], our focus diverges from learning prompt embeddings that usually depend heavily on the input. Instead, for all the images that were found to produce unreliable predictions, we try to selectively modify prompts with the use of knowledge hierarchy methods. Additionally, we keep our prompts in a natural language format rather than representing it as vector representations. This distinction showcases the unique characteristics of our zero-shot, post-hoc method in utilizing knowledge hierarchy for prompt adjustment without the need for supervised data.

2.3 Label Hierarchy

Several prior studies have explored the use of label hierarchies or label ontologies, which are essentially relational graphs among semantic labels. Notably WordNet, a widely used concept ontology, has been applied to visual recognition problems. In a study described in [19], a hierarchical graph was proposed to demonstrate the meaningful relationships that are available among the labels. Through doing this, they were able to notably enhance the object classification tasks performed by the model. Similarly in a study performed in [20], it has been showed that the semantic distance on the WordNet ontology graph can be used for knowledge transferring in zero-shot learning. These studies share similarities with our approach, leveraging label semantics encoded in such hierarchies. However, our work differs in the way that we use label hierarchies in settings where both text and images will share the same underlying latent space. This unique aspect enables us a direct exploitation of the hierarchical structure to enhance recognition patterns of the model.

3 Proposed Methodology

3.1 Background

CLIP [1], which stands for Contrastive Language-Image Pretraining, has garnered significant attention for aligning image and textual representations. This method mainly comprises of 2 parts: a visual encoder $V(x)$ for images and a language encoder $L(t)$ for text descriptions, both operating in the same d -dimensional space. The training stage involves both encoders, jointly using a contrastive loss, being applied to a large dataset containing paired images and captions. Once trained, CLIP becomes capable of zero-shot classification for downstream tasks. To classify images into C -classes, category descriptions $\{t_c\}_{c=1}^C$ are being created through prompting, by having the template such as "a photo of a {class}" [31]. Then the probability of labelling the image x is thus defined in the following way:

$$p(y|x) = \frac{\exp(\cos(V(x), L(t_y))/t)}{\sum_{c=1}^C \exp(\cos(V(x), L(t_c))/t)}$$

here t represents the temperature parameter and y , where $y \in \{1...C\}$ is our target class.

3.2 AGP: Automatic Generated Prompts for Vision-Language Models

In this paper, our goal is to enhance the performance of CLIP for both existing and new categories — referring to those with and without available data, respectively. Importantly, we aim to achieve this improvement while upholding the data efficiency demonstrated by previous approaches. For doing that, we are proposing a new approach called Automatic Generated Prompts (*AGP*) for improving the CLIP model's performance.

To ensure efficient utilization of data, we opt for Variational Autoencoders (VAE) [7], a specific type of autoencoder in unsupervised machine learning. VAEs excel in capturing and representing data in a condensed latent space, offering the capability to generate new data samples resembling the training dataset [24]. Importantly, VAEs are chosen for their potential ease of training and effectiveness, particularly in scenarios with limited data samples. The VAE architecture consists of two main parts: an encoder and a decoder. At first, the encoder E takes visual features (obtained from the CLIP visual encoder) for seen classes and converts it to a

latent code z . Subsequently, the generator takes in both this code z and the corresponding class c to reconstruct features for the image. Both the encoder E and the generator G is optimized using the following formula [32]:

$$L = L_{recon} + L_{KL}$$

where L_{recon} represents the reconstruction loss, measuring how well the generated output $G(z, c)$ approximates the original input. It is calculated as the negative log-likelihood of the generated output given the input as follows:

$$L_{recon} = -\log G(z, c)$$

and then L_{KL} represents the Kullback-Leibler divergence between the distribution produced by the encoder $E(x)$ and a predefined prior distribution $p(z|c)$:

$$L_{KL} = KL(E(x) \parallel p(z|c))$$

A generator built upon the CLIP model is also employed to harness the pre-existing knowledge of aligned vision and language representations within the pretrained CLIP. Drawing inspiration from established prompt learning techniques [25, 26, 17], we opt to create instance-specific prompts instead of directly generating features in the following manner:

$$p(z) = [p_1 + r, p_2 + r, \dots, p_L + r],$$

where z is the latent code, r is the local bias that is acquired by passing the latent code z through an artificial neural network, specifically a MLP (Multi-Layer Perceptron) and L is the length of the set of learnable prompts. The figure below describes the improvements that can be made by using instance-specific prompts in more details:

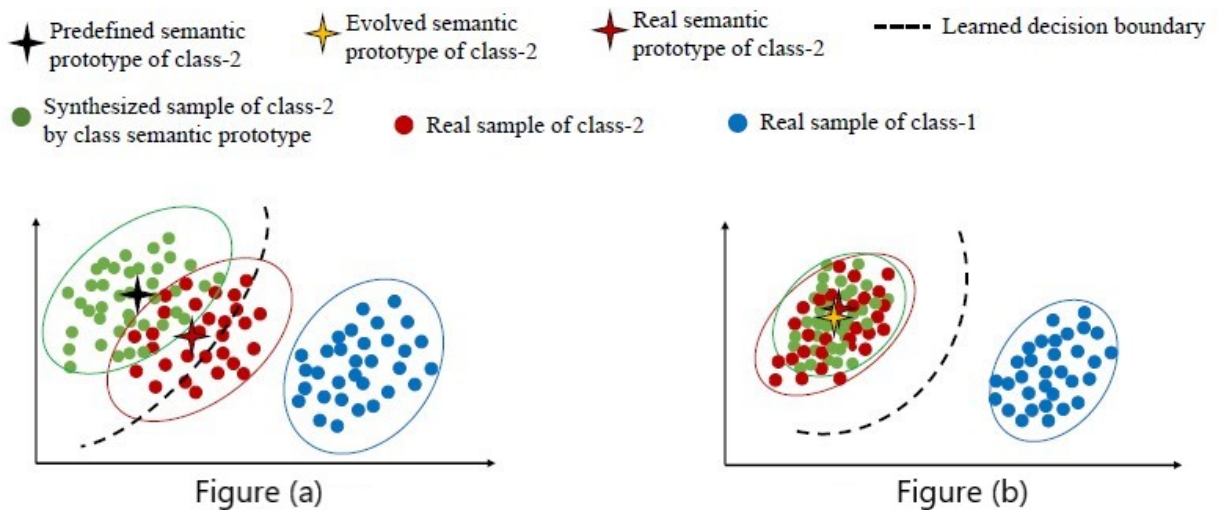


Fig. (a) illustrates the challenge of unaligned visual features synthesized by an unaligned semantic prototype. In this scenario, there is a discrepancy between the semantic attributes and the synthesized visual features, leading to inaccuracies in feature representation. However, in Fig. (b), instance-specific prompts are utilized to align visual features with the semantic prototype. Through this method, the visual features synthesized closely match the attributes specified in the semantic prototype, resulting in more accurate and aligned representations.

Then by combining these prompts with the token embedding linked to each class name, we can obtain the created features using the equation [33]:

$$m = L(t), \quad t = \{p(z), e_c\},$$

Here, L represents the fixed language encoder, while e_c denotes the token embedding for the relevant class names. In the training phase, CLIP remains frozen, and only components such as the encoder, the Multi-Layer Perceptron, and the global prompts are optimized. This approach allows us to benefit from CLIP's pretrained knowledge and generate more accurate features, especially for new classes that do not have a sufficient amount of data.

Lastly, in the tuning stage of the process, we generate features for new, unseen classes. For each new class that we come across, we will obtain its visual features of its image using the CLIP's Visual Encoder, then we convert it to the latent code and customized prompts using the prompt learning method [34]. Subsequently, by using it with the corresponding class names, we can generate the features. This approach allows us to create a new dataset for the unseen classes. When this new dataset is combined with the labelled dataset for the seen classes, a more complete dataset for all the classes shall be obtained, effectively creating a "*bridge*" between CLIP's existing knowledge and the unseen classes. Consequently, we employ existing fine-tuning techniques like CoOp [6, 17] to refine CLIP further.

3.3 Semantic Hierarchy Augmentation

In addition to the approach stated above, we use one another technique to try and improve the prediction accuracy of the CLIP model. It has been observed that for ImageNet dataset [11], there is a notable gap between zero-shot top-1 accuracy (64.2%) and top-5 accuracy (89.4%), with this model, thus indicating room for improvement. Further investigation revealed that many top-1 errors are caused by ambiguous text prompts, stemming from ImageNet's WordNet hierarchy. Notably, the ImageNet dataset lacks uniform specificity in class names. Some are overly abstract, while others are too specific; an observation highlighted by the fact that 350 classes have children, whereas the remainder do not. For instance, some names are too broad, which makes it difficult for the model to differentiate between similar but more specific objects like "hot-air balloons" and "airships." Conversely, overly specific names like "tuskers" can lead to confusion with broader categories like "Asian elephants."

As a result, we are proposing to identify images with high prediction errors and enhancing their accuracy using WordNet hierarchy. We plan on estimating prediction confidence, by introducing variations in text prompts for each image and applying various transformations to the image, such as rotations or crops, to gauge the impact on predictions [35]. For each specific change that we initialize, the prediction confidence can be calculated using common methods such as the *Maximum SoftMax Probability* [8]. This involves taking the SoftMax function's maximum output among all predicted classes. Mathematically, it can be represented as follows:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^N e^{z_j}}$$

where z_i is the is the logit for class i and N is the total number of classes. We also introduce a prediction confidence metric, considering the consistency of predictions across different prompts and image changes. This metric involves measuring the variance of predicted probabilities for a given image across its variations.

Establishing a threshold for low prediction confidence is also crucial at this stage. If the metric falls below this predefined threshold, it signals the model's uncertainty about the prediction. An iterative refinement process is implemented where the modified prompts are fed back into the model, and the prediction confidence is reassessed. Once low prediction confidence is identified,

our focus shifts to selectively modifying prompts. Here, we leverage the WordNet semantic label hierarchy by expanding the set of possible labels for the image, *i.e.*, a labelling technique that makes use of both parent labels for broader context and children labels for more specific details from WordNet. This is described in more details below:

1. Adding Ancestors (Top-Down Label Augmentation)

In CLIP, we can include broader categories or super classes as labels in addition to the specific label. This helps the model to understand the context better. For example, if the original label is "hot-air balloon," we can also include labels like "aircraft" or "vehicle" as ancestors. To do this, at first the parent node for the class is identified using WordNet. Then, we combine it with the original class as $[c; p(c)]$, where c is the class name and $p(c)$ is its parent node. This approach enhances classification accuracy by applying it to the top-5 predicted classes for each image. From this, we then can deduce the corresponding formula as follows:

$$L(x, c) = L(x, [c; p(c)])$$

Here L stands for "logit" and $L(x, c)$ denotes the logit score for a specific class c based on the input image x .

This process, as discussed more in [36], basically involves using the WordNet to identify the parent node and then combining it with the original class. This leads to a reordering of predictions for images which were initially considered unreliable. It's also worth noting that WordNet includes overly abstract class names like "entity" or "artifact," and academic terms, such as "Lepidopterous" (related to butterflies and moths). These terms are very broad and lack distinct features for accurate predictions, so we simply remove such instances. Basically, in order to ensure the accuracy and strength of our predictions, we simplify WordNet by removing words that appear less often or don't provide much useful information in the CLIP training data.

2. Adding Children (Bottom-Up Label Augmentation)

Conversely, we can include more specific labels or subclasses as labels in addition to the original one. For instance, if the label is "tusker" (a specific type of elephant), we can add labels like "African elephant" or "Asian elephant" as children. To ensure accurate classification using CLIP, we adapt our approach by augmenting class names with more

detailed child subclasses. For every class c that has different subcategories in the WordNet hierarchy, we change how we score them. Instead of just looking at the score for the main class, we also consider the scores for all its subcategories. We pick the highest score among all of them. This is expressed mathematically as:

$$L(x, c) = \text{Max} [L(x, c), L(x, c_1), L(x, c_2), \dots, L(x, c_r)]$$

here $c_1 \dots c_r$ represent the individual children related to the main class c . We apply this approach to the class names of the top-5 predicted results and reorder the top predictions accordingly.

Hence, we can enhance each class c by considering both its subcategories (children) and its more general category (parent). This process involves transferring meaningful information bidirectionally – from the more generic parent class to the specific class c and from the specific subcategories to the main class [36]. The idea is that the parent class can help clarify instances at a higher, more abstract level, while the subcategories offer detailed attributes that connect the image to the general class concept c . The end scoring formula is as follows:

$$L(x, c) = \text{Max}[L(x, [c, p(c)]), L(x, [c_1, p(c)]), \dots, L(x, [c_r, p(c)])]$$

The pseudocode for the above approach can be defined as follows:

Input: an image x for classification, a set of top-5 possible class labels C_{top5} , and the WordNet hierarchy H

Output: the class identified for x

1. Initialize an empty set C
 2. For each class c in C_{top5} :
 - a. Add the pair $(c, \text{parent}(c))$ to set C (Top-down augmentation)
 3. For each class c in C_{top5} with at least one child:
 - a. For each child c_j of c :
 - i. Add the pair $(c_j, \text{parent}(c))$ to set C (Bottom-up augmentation)
 4. Find the class with the highest logit score from set C and assign it to c_{max}
 5. If the predicted class c_{max} is in C_{top5} :
 - a. Set the final prediction as c_{max}
- Else:
- a. Set the final prediction as the parent of c_{max}

4 Experiments and Results

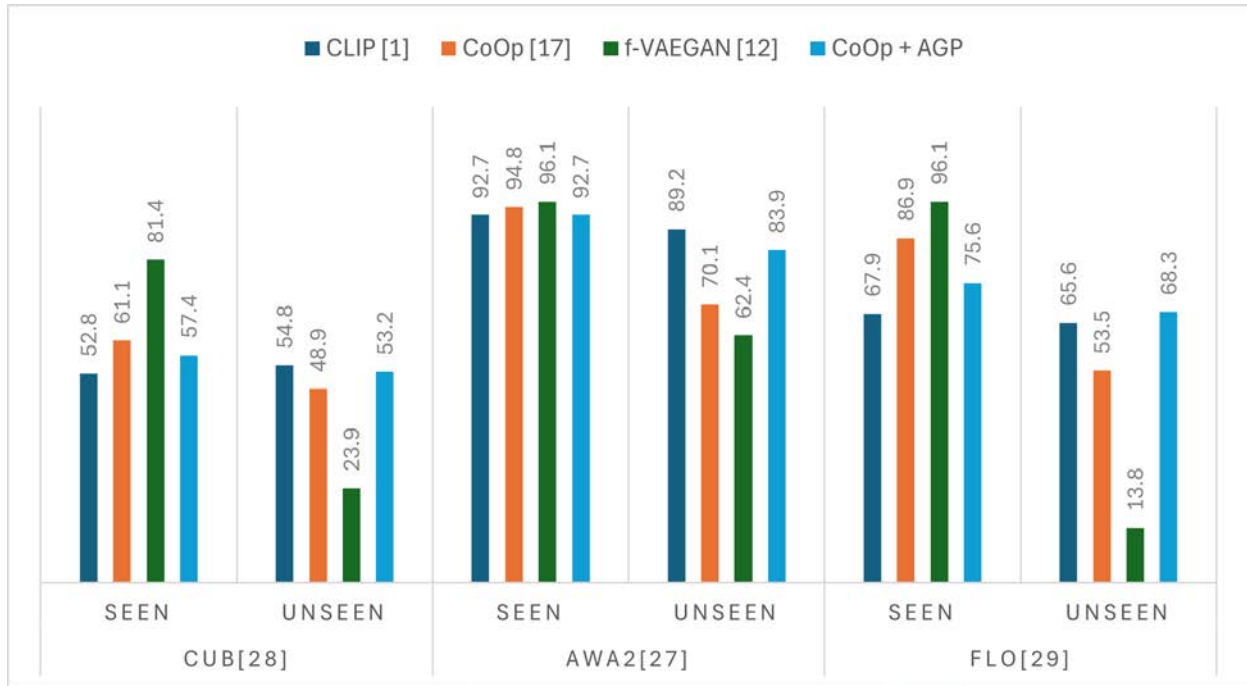
4.1 Generalized zero-shot learning

In our experimental approach for the *AGP* system, we evaluate our method for the generalized zero-shot classification tasks. For this instance, we follow the same experimental guidelines as specified in [27]. Our experiments focus on three well-established datasets for zero-shot recognition: Animals with Attributes2 [27] (AWA2), Oxford Flowers [29] (FLO), and Caltech-UCSD-Birds [28] (CUB). To ensure fairness and accuracy in evaluation, similar data splitting techniques are used as in [27]. Additionally, we use f-VAEGAN [12] as our baseline method, which will provide us with a benchmark against which our proposed approach can be evaluated for effectiveness in zero-shot learning tasks.

In order to implement all of this, we are using a *VAE* encoder, a lightweight *MLP*, and the available *CLIP* model. The *MLP* and the *VAE* encoder are basic neural networks with 2 layers, having 4,096 hidden units and activation functions that promote non-linearity, such as a sigmoid function. We use ViT-B/16 [21] as the image encoder and a transformer [23] as the text encoder for *CLIP*. These encoders come with pretrained weights, and we maintain their fixed state throughout the training process. In order to be consistent with *CLIP*'s guidelines [1], we also ensure uniformity by scaling the features to fit within a unit sphere and employ Mean Squared Error (*MSE*) as the *VAE*'s reconstruction loss. The training of all networks is supported by the SGD optimizer, by maintaining a suitable learning rate.

During training phase, we introduce variety by randomly creating a batch of features for new classes by using the *CLIP*'s visual encoder to capture image features and the associated class attribute that is derived from our previous template of "a photo of a {class}". These newly generated features are then mixed with the original batch of data, forming a refreshed training batch. Similar to previous studies [27], we also calculate the average accuracy for each class, while considering both known and unknown classes. Lastly, in the fine-tuning phase of *CLIP*, we take a practical and readily available approach, implementing methods such as *CoOp* established in existing research papers [5, 17, 25, 17].

The results of our experiment for the generalized zero-shot learning task are shown on the table below. As mentioned above, the experiment was carried out using 3 datasets which are CUB [28], AWA2 [27], and FLO [29]. The models were trained using data from known classes and then tested on a combination of both known and unknown classes. The following results are presented as the average top-1 accuracy for both known and unknown classes:



From the above results, we observe that when compared with CLIP, the fine-tuning implemented by CoOp [17] showed improvements in seen class performance. We can see that for the seen classes, the fine-tuning leads to a performance increase of +8.3%, +2.1% and +19% respectively on the CUB, AWA2 and the FLO datasets. But we can also notice that it struggled with unseen classes, which is evident from the performance decline, thus indicating potential overfitting. The proposed method of CoOp + AGP however seems to be addressing this by leveraging the generative prompts technique. As evident, we can see improvements of +4.3%, +13.8% and +14.8% on unseen classes as compared to CoOp. The findings also reveal that f-VAEGAN [12] has the best performance on seen classes, but its performance diminishes notably when dealing with unseen ones, implying that there is a pronounced tendency for such models to overfit the seen classes data.

Qualitative Evaluation

We also conducted a thorough qualitative evaluation to better understand the effectiveness of our AGP approach when integrated into the baseline CoOp framework. Leveraging t-SNE (t-distributed Stochastic Neighbor Embedding) visualization [37], a powerful tool for dimensionality reduction and visualization, we investigated both the real and synthesized sample features. Specifically, we randomly selected ten classes from the CUB dataset to ensure representative evaluation. On the figure below, the visualization showcased on the left side depicts sample features synthesized solely by CoOp. Notably, there's a discernible deviation between these synthesized features and the real ones, highlighting a discrepancy or misalignment in the visual and textual representations of the classes. Contrastingly, the visualization on the right, depicting features synthesized by CoOp+AGP, suggests an improved performance in prediction accuracies of unknown classes. This visualization shows that our AGP approach can greatly help generative ZSL methods to create synthesised features that can closely match the real sample features.

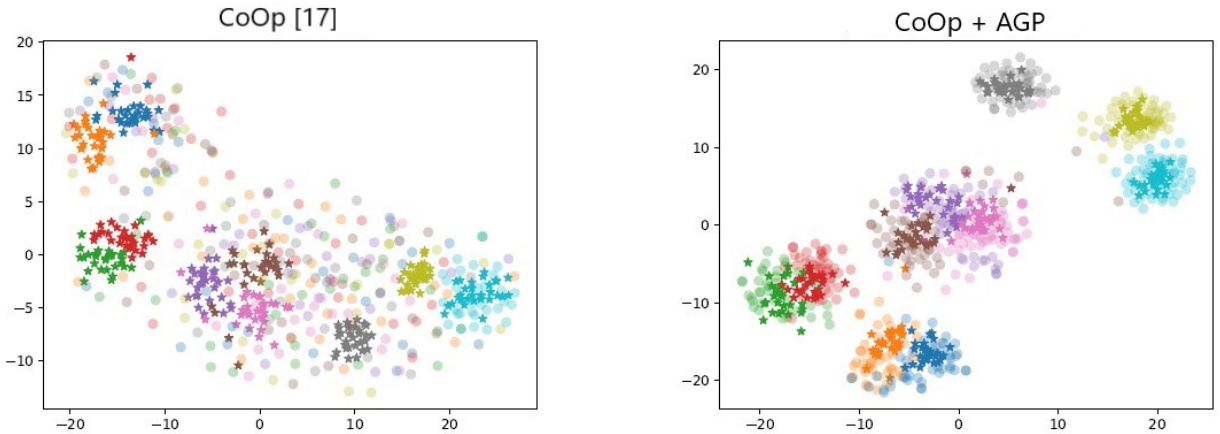


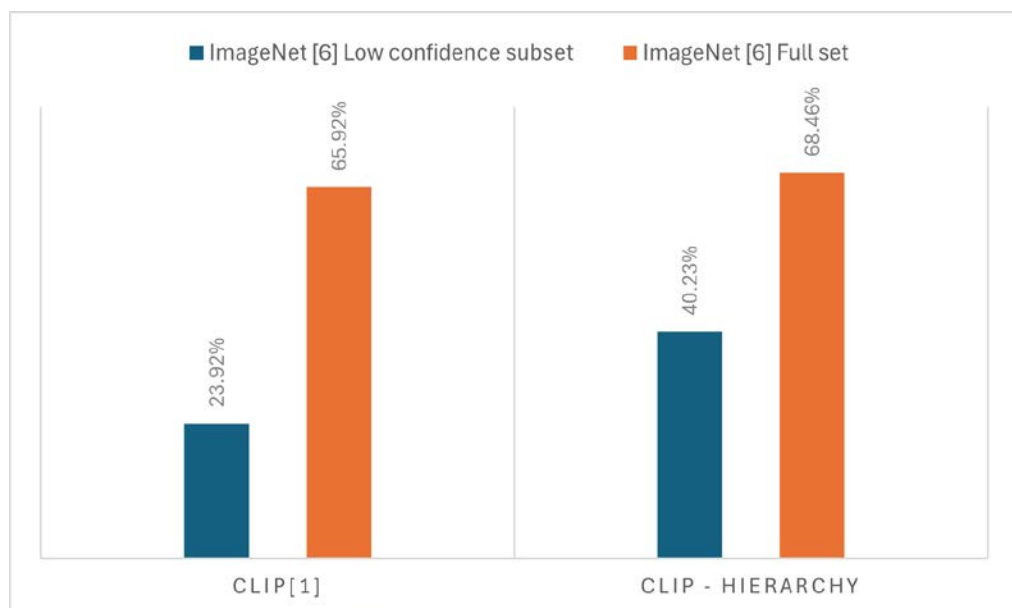
Fig. Qualitative evaluation with t-SNE visualization. We represent randomly selected 10 classes from CUB dataset using 10 different colors. Real sample features are denoted by ○, while synthesized sample features are denoted by ★.

4.2 Enhancing ImageNet Classification with Semantic Hierarchy Augmentation

Additionally, while utilizing semantic hierarchy augmentation, we observe some improvements in accuracy, particularly for images with low confidence predictions. To accomplish this, we employed both top-down and bottom-up methods for augmenting labels, focusing specifically on the subset of images with low confidence scores. For a more effective combination of child and parent names, a prompt template has been devised in order to create a new class name c in plain language format. It's structured as "{child} which is a kind of {parent}".

In order to identify the cases with low prediction accuracies, we employ the Maximum SoftMax Probability [8] technique as a confidence estimation method. This involves calculating the SoftMax probability for each class and selecting the maximum value as the confidence score. By applying this method across different text prompts and image transformations, we can obtain a measure of consistency in predictions. If the predictions are inconsistent, it indicates low prediction confidence. Once we identify the set of images with low confidence scores, the top-down and bottom-up labelling techniques are implemented specifically on these images to improve their classification accuracy [36].

Using the semantic hierarchy augmentation technique, we are able to see some notable improvements as well. The table below showcases a notable 16.31% improvement in top-1 accuracy for the low confidence subset (from 23.92% to 40.23%). This improvement extended to an overall 2.54% increase in top-1 accuracy for all samples in ImageNet (from 65.92% to 68.46%).



4.3 Ablation Study

Using different instances of generative prompts

As described earlier in the methodology section of our AGP approach, by using some of the available prompt learning techniques [25, 26, 17], we are creating prompts that consist of two parts: fixed global prompts $(p_1, p_2, \dots, p_L)$ and a local bias that is tailored to each instance (r) . These are combined by adding the bias to each global prompt, like this: $p = [p_1 + r, p_2 + r, \dots, p_L + r]$. We now explore how using different prompt structures can affect the overall performance of predicting the unseen classes. The term "global" is used to refer to using only global prompts $[p_1, p_2, \dots, p_L]$, while "sequential" means using the same or sequentially applied local biases, like $[r, r, \dots, r]$ or $[r_1, r_2, \dots, r_L]$ respectively. The result in the table below indicates that employing global prompts with consistent local bias for each instance yields the best performance. However, relying solely on local prompts, regardless of their sequence, negatively impacts the performance of new classes. It could be due to the absence of global prompts failing to retrieve and use important information about the classes. Interestingly enough, using both global and sequential prompts leads to the worst performance, possibly due to training instability as the model is leaning from both the global and sequential prompts, so it may be struggling to find the right balance between capturing general trends and adapting to individual instances.

global	sequential		Seen	Unseen
Used	Used		81.84	71.76
Not Used	Used		83.12	72.32
Used	Not Used		79.97	74.55
Not Used	Not Used		80.81	72.68

Fig. The table shows the average results evaluated on the datasets of CUB [28], AWA2 [27], and FLO [29]. 'global' indicates whether global prompts are utilized, while 'sequential' indicates whether the local bias follows a sequential or identical pattern.

Using different backbones

To also explore how well our semantic hierarchy augmentation approach works with different model architectures and sizes, we tested it with two additional backbones of CLIP: ResNet-50 and ResNet-101, which are based on convolutional neural networks (CNNs). The results, shown in the table below, indicate that our method improves accuracy across different CLIP models on the ImageNet dataset [6].

Backbone	ResNet-50	ResNet-101
Low confidence subset	+13.78%	+11.89%
Full set	+4.85%	+3.59%

5 Conclusion, Limitations and Future Work

In conclusion, our research which endeavours in the realm of vision-language models have yielded promising advancements through the introduction of the *AGP* methodology and Semantic Hierarchy Augmentation. *AGP*, a generative approach, adeptly tackles the challenge of fine-tuning pretrained models for downstream tasks with limited labelled data, particularly in scenarios where certain classes lack relevant data. This is achieved by generating features for new, unseen categories solely based on their class names. Concurrently, Semantic Hierarchy Augmentation capitalizes on the hierarchical structure of WordNet to enhance the robustness of zero-shot classification, particularly for images with low-confidence predictions. We were able to show performance improvements using both of these approaches on a range of datasets and also conducted ablation studies to better understand the individual contributions of each of the techniques used. However, despite these promising advancements, there are still several avenues for future exploration and improvement.

As deep learning models like CLIP become more prevalent in real-world applications, the need for interpretability and transparency in their decision-making processes becomes paramount. While *AGP* and Semantic Hierarchy Augmentation may enhance the performance of such vision-language models, they may lack explicit interpretability, making it challenging to understand why certain classifications are made. This lack of transparency could hinder trust and usability, especially in critical applications where decision rationale is crucial, such as medical diagnosis or autonomous driving.

To address this limitation, future work could focus on integrating interpretability techniques directly into the *AGP* framework. For instance, attention mechanisms [38] could be employed to highlight which image regions or textual prompts are most influential in driving the model's predictions. By visualizing these attention maps, users can gain insights into the model's reasoning process, understanding which features or concepts are being prioritized during classification. Similarly, saliency maps [39] could highlight the most relevant regions of an image or the most informative words in a textual prompt, providing further transparency into the decision-making process.

6 References

1. A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al. Learning transferable visual models from natural language supervision. In *ICML*, 2021.
2. J.B. Alayrac, J. Donahue, P. Luc, A. Miech, I. Barr, Y. Hasson, K. Lenc, A. Mensch, K. Millican, M. Reynolds, et al. Flamingo: a visual language model for few-shot learning. *arXiv:2204.14198*, 2022.
3. P. Liu, W. Yuan, J. Fu, Z. Jiang, H. Hayashi, and G. Neubig. Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing. *ACM Computing Surveys*, 55(9):1–35, 2023.
4. F. Wang, M. Li, X. Lin, H. Lv, A. G. Schwing, and H. Ji. Learning to decompose visual features with latent textual prompts. In *ICLR*, 2023.
5. R. Zhang, W. Zhang, R. Fang, P. Gao, K. Li, J. Dai, Y. Qiao, and H. Li. Tip-adapter: Training-free adaption of clip for few-shot classification. In *ECCV*, 2022.
6. Y. Lu, J. Liu, Y. Zhang, Y. Liu, and X. Tian. Prompt distribution learning. In *CVPR*, 2022.
7. D. P. Kingma and M. Welling. Auto-encoding variational bayes. In *ICLR*, 2014.
8. D. Hendrycks and K. Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. *arXiv preprint arXiv:1610.02136*, 2016.
9. J. Deng, W. Dong, R. Socher, L. J. Li, K. Li, and L. Fei-Fei. Imagenet: A large-scale hierarchical image database. In *IEEE conference on computer vision and pattern recognition*, pages 248–255, 2009.
10. T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean. Distributed representations of words and phrases and their compositionality. In *NeurIPS*, 2013.
11. W. Xu, Y. Xian, J. Wang, B. Schiele, and Z. Akata. Attribute prototype network for zero-shot learning. In *NeurIPS*, 2020.
12. Y. Xian, S. Sharma, B. Schiele, and Z. Akata. A feature generating framework for any-shot learning. In *CVPR*, 2019.
13. Y. Xian, T. Lorenz, B. Schiele, and Z. Akata. Feature generating networks for zero-shot learning. In *CVPR*, 2018.
14. R. Felix, I. Reid, G. Carneiro, et al. Multi-modal cycle-consistent generalized zero-shot learning. In *ECCV*, 2018.

15. G. S. Xie, L. Liu, X. Jin, F. Zhu, Z. Zhang, J. Qin, Y. Yao, and L. Shao. Attentive region embedding network for zero-shot learning. In *CVPR*, 2019.
16. X. Zhai, X. Wang, B. Mustafa, A. Steiner, D. Keysers, A. Kolesnikov, and L. Beyer. Lit: Zero-shot transfer with locked-image text tuning. In *CVPR*, pages 18123–18133, 2022.
17. K. Zhou, J. Yang, C. C. Loy, and Z. Liu. Learning to prompt for vision-language models. *IJCV*, 2022.
18. M. Shu, W. Nie, D. Huang, Z. Yu, T. Goldstein, A. Anandkumar, and C. Xiao. Test-time prompt tuning for zero-shot generalization in vision-language models. *arXiv preprint arXiv:2209.07511*, 2022.
19. J. Deng, N. Ding, Y. Jia, A. Frome, K. Murphy, S. Bengio, Y. Li, H. Neven, and H. Adam. Large-scale object classification using label relation graphs. In *ECCV*, pages 48–64, 2014.
20. M. Rohrbach, M. Stark, G. Szarvas, I. Gurevych, and B. Schiele. What Helps Where – And Why? Semantic Relatedness for Knowledge Transfer. In *CVPR*, pages 910–917, 2010.
21. A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al. An image is worth 16x16 words: Trans-formers for image recognition at scale. In *ICLR*, 2021.
22. K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *CVPR*, 2016.
23. A. Vaswani, N. Shazeer, N. Parmar, J. Uszko-reit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. In *NeurIPS*, 2017.
24. D. P. Kingma and M. Welling. Auto-encoding variational bayes. In *ICLR*, 2014.
25. K. Zhou, J. Yang, C. C. Loy, and Z. Liu. Conditional prompt learning for vision-language models. In *CVPR*, 2022.
26. Y. Lu, J. Liu, Y. Zhang, Y. Liu, and X. Tian. Prompt distribution learning. In *CVPR*, 2022.
27. Y. Xian, C. H. Lampert, B. Schiele, and Z. Akata. Zero-shot learning—a comprehensive evaluation of the good, the bad and the ugly. *IEEE TPAMI*, 41(9):2251–2265, 2018.
28. P. Welinder, S. Branson, T. Mita, C. Wah, F. Schroff, S. Belongie, and P. Perona. Caltech-ucsd birds 200. Technical Report, 2010.
29. M. E. Nilsback and A. Zisserman. Automated flower classification over a large number of classes. In *ICVGIP*, 2008.

30. S. Chen, G. S. Xie, Y. Liu, Q. Peng, B. Sun, H. Li, X. You, L. Shao. HSVA: Hierarchical Semantic-Visual Adaptation for Zero-Shot Learning. *arXiv:2109.15163*, 2021.
31. Y. Lu, J. Liu, Y. Zhang, Y. Liu, X. Tian. Prompt Distribution Learning. *arXiv:2205.03340*, 2022.
32. J. Kim, J. Kong, J. Son. Conditional Variational Autoencoder with Adversarial Learning for End-to-End Text-to-Speech. In *ICML*, 2021.
33. J. Gong, H. Eldardiry. Prompt-based Zero-shot Relation Extraction with Semantic Knowledge Augmentation. *arXiv:2112.04539*, 2021
34. J. Guo, C. Wang, Y. Wu, E. Zhang, K. Wang, X. Xu, S. Song, H. Shi, G. Huang. Zero-Shot Generative Model Adaptation via Image-Specific Prompt Learning. In *IEEE/CVF*, 2023 pp. 11494-11503.
35. B. An, S. Zhu, M. A. Panaitescu-Liess, C. K. Mummadi, F. Huang. More Context, Less Distraction: Zero-shot Visual Classification by Inferring and Conditioning on Contextual Attributes. In *ICLR*, 2023.
36. Z. Novack, J. McAuley, Z. C. Lipton, S. Garg. CHiLS: Zero-Shot Image Classification with Hierarchical Label Sets. In *ICML*, 2023.
37. L. V. D. Maaten, G. E. Hinton. Visualizing data using t-sne. *Journal of Machine Learning Research*, 9:2579–2605, 2008.
38. R. Wang, Z. Chen, C. Chen, J. Ma, H. Lu, X. Lin. Compositional Text-to-Image Synthesis with Attention Map Control of Diffusion Models. *arXiv:2305.13921*, 2023
39. J. Sigut, F. Fumero, J. Estévez, S. Alayón, & T. Díaz-Alemán. In-Depth Evaluation of Saliency Maps for Interpreting Convolutional Neural Network Decisions in the Diagnosis of Glaucoma Based on Fundus Imaging. *Sensors*. 24. 239.10.3390/s24010239, 2023.